Element E

STEM Elements		Proposed solutions in design:
Engineering	The principles of engineering were applied to our design as we conducted our research. Math and science were vital in making our final deciding factors. We used biological sciences when determining which materials were safe and nontoxic. Physical sciences were used when researching how students with motor issues would interact with the design.	Engineering is everywhere and used in almost every situation. Optimizing any part of our cause and effect device took knowledge and application of engineering principles.
Aesthetics	The aesthetic of the device is vital because if it looks dull and boring, the students are less likely to engage. While it may not impact the processing aspect, keeping your design bright with vivid colors will positively impact the amount the children willingly interact.	Our design has an engaging look due to the fact that we will utilize textures, colors, and materials to encourage interaction.
Non/Mouth-Safe Materials (Science)	**Unfinished Wood:** Wood is commonly used in the construction of buildings and designs involving carpentry. - Some wooded plants are toxic, you wouldn't want to grab it as a resource and allow it to be in the reach of children - Woods are treated with insecticides and chemicals to prevent rotting - Woods are prone to splitting - Porous, therefore able to soften when wet and absorb unsanitary elements **Toxic Paint:** Paint is used to create color and aesthetic for buildings and designs - Paint contains VOC (volatile organic compounds) - Possibly cause irritation to the eyes, skin, and respiratory tract - Acrylic paints typically have significantly less VOCs than enamel and latex paints **Glue**: Glue is compound used to stick things together - Hexane and Toluene - has health effects ranging	We are using science to determine which materials are non toxic and safe for our stakeholders. We are utilizing paint made with plant-based dyes that have low or no VOCs, as well as covering a majority of our design with silicone; a mouth safe option for young students.

	from irritation to the eyes, nose, and throat to nerve damage, unconsciousness, and death **Soft Plastic:** any type of plastic that you can scrunch in your hand - plastic toys commonly contain phthalates	
Required Distancing (math and measurements)	Motor skills involve manipulation, such as grasping, threading and buttoning. Motor disabilities are a possible attribute of our stakeholders. To accommodate this, we used mathematical procedures to determine what the most ideal spacing for the separate buttons on the design to be. - Distance from the device decreases accuracy, the further the buttons are apart the easier the device is to use. The weight of the device needs to be portable and convenient. To define the correct weight and dimension of the device we used concepts such as kinetic Friction versus the gravitational pull of 9.8 (m/s2).	In our design we are using the 2 inch-spaced buttons, this is the ideal distance to ensure accuracy from the students. The ideal weight for the device is 10lb, using equations of weight for the materials used, we fit within the 5-10 lbs constraint.
Chemistry	Chemistry can be used to decide which materials are safe to utilize in a children's inactive device. Assuming the device may enter a student's mouth, researching phthalates; a class of chemicals used to soften plastic, is a key element to look into. Phthalates are known to cause adverse health effects such as hormone disruptions, developmental problems and asthma. Common materials that contain chemicals as such will be avoided. This ensures the safety of the students using our device.	We used the principles of chemistry to decide which materials were safe to utilize in our design, avoiding any toxic options.
Technology	Technology can be used to create the light, sound, and vibration rewards that are featured in the device. Coding tells the design which actions to perform and how to complete tasks.	We are going to code with Arduino/raspberry pie to add our stimulating reward responses.
Geometry	The aspects of geometry are essential to incorporate into our design before the building process begins. Form factoring can	We are going to utilize geometry to be able to predict how the separate pieces of our device

	be used during the sketching/OnShape processes to ensure that all of our programming and additional constructional pieces will fit comfortably with one another when the physical design is built.	will eventually fit together.

The expert that was able to provide our group with feedback, on our STEM principles, was teacher Eric Scheidly. We introduced our STEM research to him during an educational meeting on Wednesday, February 7th, 2024. Mr. Scheidly received a degree in psychics from Drexel University. He teaches academic, honors and AP level psychics and electronic courses at Plymouth Whitemarsh High School, as well as being a teacher in psychics at Drexel University. During the group's conversation with him, we received feedback on an additional STEM principle that can be added and utilized; Mr. Schieldy also approved the group's previously created principles. Mr. Scheidly urged us to incorporate pre-planning the geometry/form factoring of our design. He was able to provide insight on past projects, where previous young engineers did not plan out how they were going to incorporate the innards of their device. This lack of planning later created conflict, when things weren't able to fit into their design correctly, with this information the group hoped to not repeat past mistakes.

Following our meeting, we began our research on the form factoring for our cause and effect device. We took into consideration the size of the physical system we are planning to use, this was previously not something we had examined. Originally planning on using ARDUINO, our research concluded that a program system like Raspberry Pie was smaller as well as being more direct to the tasks we need to complete with our design.